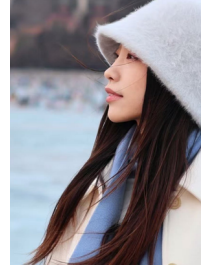


YIJU GUO

Resumé

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EDUCATION

Renmin University of China

Sep. 2022 – Present

Ph.D. in Artificial Intelligence, School of Artificial Intelligence | GPA: 3.94/4.0

Beijing, China

Advisor: Prof. Yankai Lin | Co-advised by Prof. Zhiyuan Liu (THUNLP, Tsinghua University)

Research focus: Deep learning, reinforcement learning, model distillation for LLMs and LLM-based agent systems.

Dalian University of Technology

Sep. 2017 – Jul. 2022

B.Eng. in Digital Media Technology | GPA: 4.02/5.0 (90.02/100) | Rank: 1/99 (Comprehensive), 2/99 (Academic)

Dalian, China

Honors: National Scholarship ($\times 2$), Outstanding Graduate of Dalian, First-Class Academic Scholarship ($\times 3$)

Relevant Coursework: Computer Vision (100), Advanced Programming (98), Signal Processing (98), Discrete Mathematics (97), Probability Theory (95)

RESEARCH & INDUSTRY EXPERIENCE

LENS: Instruction Purification for RL Reasoning (ACL 2026)

Jul. 2025 – Jan. 2026

First Author | THUNLP & RUCBM | Baidu Search, Research Intern | Supervisor: Yankai Lin

Beijing, China

- **Reinforcement Learning with Verifiable Rewards (RLVR)** has advanced LLM reasoning but remains constrained by **inefficient exploration under limited rollout budgets**, leading to low sampling success and unstable training. Discovered that many exploration failures arise not from problem difficulty, but from a small number of prompt tokens that **introduce interference**.
- Proposed the **Less Noise Sampling Framework (LENS)**, an **online selective rollout** framework: **Stage 1** identifies and removes interference tokens within low-success prompts via **Interference Score**, producing purified prompts that yield a higher proportion of successful rollouts; **Stage 2** transfers successful rollouts from the purification process to supervise **policy optimization on the original noisy prompts**, enabling the model to learn to ignore interference in real-world noisy prompting settings.
- LENS significantly outperforms **GRPO**, achieving a **Pareto improvement** in performance–efficiency trade-offs, with an average **+3.88%** gain and over **1.6 $\times$** faster convergence across **7 math reasoning benchmarks**. Also surpasses both **scaling exploration** (GRPO with doubled rollouts) and **zero-variance prompt filtering** baselines (DAPO and GRESO) while using substantially fewer computational resources. Evaluated on Llama-3.2, Qwen-2.5/3 and Qwen-3 model families.

LeaF: Causal Attention Distillation Framework (NeurIPS 2025)

May 2024 – May 2025

First Author | THUNLP & RUCBM | Modelbest Inc., Research Intern | Supervisor: Yankai Lin

Beijing, China

- Proposed **Learning to Focus (LeaF)**, a two-stage framework that reduces the influence of distracting tokens to improve LLM reasoning accuracy and generation quality in complex tasks.
- **Confounding Token Detection**: LeaF identifies confounding tokens via teacher–student **gradient-based comparisons** on sensitivity discrepancies, effectively pruning interfering tokens from training corpora and generating high-quality **counterfactual samples** through **span pruning**.
- **Causal Attention Distillation**: By contrasting model outputs on original and counterfactual samples through a **hybrid distillation loss** (standard KL divergence + counterfactual KL divergence), LeaF guides the student model to learn genuine **causal dependencies**, thereby improving reasoning capability.
- LLaMA-1B/3B-LeaF and Qwen2.5-1.5B-LeaF outperformed standard distillation on **math reasoning** (GSM8K, MATH, OlympiadBench) by **+2.41%**, on code generation (HumanEval+, LeetCode, LiveCodeBench) by **+2.48%**, and on multi-hop reasoning (2WikiMQA, Musique, HotpotQA) by **+3.24%**.
- Interpretability experiments confirm that LeaF enhances model attention to critical information while reducing reliance on irrelevant tokens, improving model reasoning capability.

CPO: Controllable Multi-Objective Alignment (EMNLP 2024, 116 citations)

Sep. 2023 – Apr. 2024

First Author | THUNLP & RUCBM | Tencent WeChat, Research Intern | Supervisor: Yankai Lin

Beijing, China

- Addressed the conflict problem in multi-objective alignment for LLMs. Innovatively introduced multi-objective **preference tokens** to specify optimization directions (e.g., helpfulness, honesty), transforming multi-objective optimization into a **conditional multi-objective optimization problem**, enabling controllable optimization over different objectives and mitigating the “alignment tax” issue.
- Proposed **Controllable Preference Optimization (CPO)**, consisting of two stages: **Controllable Preference Supervised Fine-Tuning (CPSFT)** and **Controllable Direct Preference Optimization (CDPO)**. CPSFT controls generation quality via preference tokens; CDPO maximizes preferences along selected dimensions while constraining specific dimensions, effectively reducing inter-objective conflicts.

- Experiments demonstrate that CPO surpasses traditional methods (SFT, PPO, DPO, Curry-DPO) across all three objectives (helpfulness, honesty, harmlessness), achieving dual improvements in both controllability and overall performance. Also introduced the **UltraSafety** dataset, addressing safety data gaps in UltraFeedback.

SELECTED PUBLICATIONS

(1). **Yiju Guo**, Tianyi Hu, Zexu Sun, Yankai Lin. "Less Noise, More Voice: Reinforcement Learning for Reasoning via Instruction Purification." **In ACL 2026**.

(2). **Yiju Guo**, Wenkai Yang, Zexu Sun, Ning Ding, Zhiyuan Liu, Yankai Lin. "Learning to Focus: Causal Attention Distillation via Gradient-Guided Token Pruning." **In NeurIPS 2025**.

(3). **Yiju Guo***, Ganqu Cui*, et al. "Controllable Preference Optimization: Toward Controllable Multi-Objective Alignment." **In EMNLP 2024**. (116 citations)

(4). Zexu Sun, **Yiju Guo**, Yankai Lin, et al. "Uncertainty and Influence Aware Reward Model Refinement for RLHF." **In ICLR 2025**.

(5). Wenkai Yang, Weijie Liu, Ruobing Xie, **Yiju Guo**, et al. "LaSeR: RL with Last-Token Self-Rewarding." **In ICLR 2026**.

(6). Shengda Fan*, Xuyan Ye*, Yupeng Huo, Zhi-Yuan Chen, **Yiju Guo**, et al. "AgentProcessBench: Diagnosing Step-Level Process Quality in Tool-Using Agents."

TECHNICAL SKILLS

Programming:	Python (primary), C/C++
Deep Learning Frameworks:	PyTorch, Hugging Face Transformers, DeepSpeed, Megatron-LM
Model Training & Optimization:	SFT, Knowledge Distillation, RLHF/RLAIF (PPO, GRPO, DPO); Attention mechanism optimization
Distributed Systems:	Multi-GPU distributed training (VeRL, OpenRLHF), Linux, Git, Docker
Data Engineering:	Large-scale training data construction, preference data annotation (UltraSafety), agent process reward data annotation (AgentProcessBench), and pipeline development

AWARDS & COMPETITIONS

2020, 2021	National Scholarship (×2), Ministry of Education of China
2022	Outstanding Graduate of Dalian , Dalian Municipal Government
2019–2021	First-Class Academic Scholarship (×3), Innovation Scholarship (×3)
2020	MatherCup National Mathematical Modeling Challenge, First Prize (National Level)
2020	National College Mathematical Modeling Competition, Second Prize (National Level)
2019, 2021	MCM/ICM (International Mathematical Modeling Contest), Honorable Mention (×2)